

4. ~~$E(X_i - \mu)^2$~~ i.i.d, $E(X_i - \mu)^2 = \sigma^2$

$$\therefore \frac{\sum_{i=1}^n (X_i - \mu)^2}{n} \xrightarrow{P} \sigma^2$$

5. ~~不妨假设 $E X_i$ 都为 0, 否则用 $X_i - E X_i$ 代替 X_i 不是问题.~~

~~则 $r(X_i, X_j) = \frac{E X_i X_j - E X_i E X_j}{\sqrt{E X_i^2} \sqrt{E X_j^2}}$~~

由 Chebyshev 不等式, $P\left(\left|\frac{1}{n} \sum_{k=1}^n X_k - \frac{1}{n} \sum_{k=1}^n E X_k\right| \geq \varepsilon\right) \leq \frac{D \frac{\sum_{k=1}^n X_k}{n}}{\varepsilon^2} = \frac{D \sum_{k=1}^n X_k}{n^2 \varepsilon^2}$

只需证 $\lim_{n \rightarrow \infty} \frac{D \sum_{k=1}^n X_k}{n^2} = 0$

$$\text{Cov}(X_i, X_j) = \sqrt{D X_i} \sqrt{D X_j} r(X_i, X_j) \leq C \lim_{|i-j| \rightarrow \infty} \text{Cov}(X_i, X_j) = 0$$

$$\frac{1}{n^2} D \sum_{k=1}^n X_k = \frac{1}{n^2} \left(\sum_{k=1}^n D X_k + \sum_{i \neq j} \text{Cov}(X_i, X_j) \right) \quad \frac{1}{n^2} \sum_{k=1}^n D X_k \leq \frac{C n}{n^2} \rightarrow 0$$

对 $\forall \delta > 0$, $\exists N$, 当 $|i-j| \geq N$ 时, $|\text{Cov}(X_i, X_j)| < \delta$

$$\begin{aligned} \therefore \text{又对 } n > \frac{2}{\delta}, \quad \frac{1}{n^2} \sum_{i \neq j} \text{Cov}(X_i, X_j) &= \frac{1}{n^2} \left(\sum_{|i-j| \leq N} \text{Cov}(X_i, X_j) + \sum_{|i-j| > N} \text{Cov}(X_i, X_j) \right) \\ \text{满足 } |i-j| \leq N \text{ 的数对有 } 2nN &\leq \frac{1}{n^2} (2nN^2) C + (n(n-1) - (2nN - N^2)) \delta \end{aligned}$$

令 $n \rightarrow +\infty$, 由 δ 的任意性, 得该式趋近于 0

原式得证

9. 记 $X_i = \cos(iX)$

则 $EX_i = 0$, $DX_i = \frac{1}{n}$ ~~且~~ $Cov(X_i, X_j) = 0$

由第5题结论, 可知 $\frac{1}{n} \sum_{k=1}^n X_k - \frac{1}{n} \sum_{k=1}^n EX_k \xrightarrow{P} 0$

$$\therefore \frac{S_n}{n} \xrightarrow{P} 0$$

11. 设 $X_i = \begin{cases} 1 & \text{考第 } i \text{ 题} \\ 0 & \text{其他} \end{cases}$ 在 n 个题中

则 $P(X_i = 1) = \frac{1}{n}$ ~~则~~ $X_i \sim \begin{pmatrix} \frac{1}{n} & 0 \\ \frac{1}{n} & 1 - \frac{1}{n} \end{pmatrix}$ $S_n = \sum_{i=1}^n X_i$

$$EX_i = \frac{1}{n} \quad DX_i = \frac{n-1}{n^2}$$

$$Cov(X_i, X_j) = \frac{1}{n^2(n-1)}$$

$$\therefore DS_n = D\sum_{i=1}^n X_i = \sum_{i=1}^n DX_i + 2 \sum_{1 \leq i < j \leq n} Cov(X_i, X_j) = 1$$

$$\therefore P\left(\left|\frac{S_n - ES_n}{n}\right| \geq \varepsilon\right) \leq \frac{P(S_n \geq n\varepsilon)}{\varepsilon^2} = \frac{1}{n^2\varepsilon^2} \rightarrow 0$$

$$\therefore \frac{S_n - ES_n}{n} \xrightarrow{P} 0$$

$$4. E\bar{X} = E \frac{X_1 + \dots + X_n}{n} = E X_1 = 0$$

$$Var(\bar{X}) = Var \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n^2} \cdot n Var X_1 = \frac{1}{3n}$$

10.

$$\sum_{i < j} (X_i - X_j)^2 = (n-1) \sum_{i=1}^n X_i^2 - 2 \sum_{i < j} X_i X_j$$

$$= n \sum_{i=1}^n X_i^2 - \left(\sum_{i=1}^n X_i \right)^2$$

$$= n \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\therefore \frac{1}{n(n-1)} \sum_{i < j} (X_i - X_j)^2 = S^2$$

24. ~~24.1~~

$$(1) P(X_{(6)} > 10) = 1 - P(X_{(6)} \leq 10) = 1 - (P(X_1 \leq 10))^{16} = 1 - (\Phi(1))^{16}$$

$$(2) P(X_{(6)} > 5) = (P(X_1 > 5))^{16} = (1 - \Phi(-1.5))^{16} = (\Phi(1.5))^{16}$$

$$24. (1) f_{F(X)}(\eta) = \begin{cases} 1 & 0 \leq \eta \leq 1 \\ 0 & \text{其他} \end{cases} \quad \therefore F(X) \sim U(0, 1)$$

$\therefore F(X_{(6)})$ 为取自 $U(0, 1)$ 的次序统计量, 有 $F(X_{(6)}) \sim Be(6, 5)$

$$\therefore E(F(X_{(6)})) = \frac{6}{11} \quad Var(F(X_{(6)})) = \frac{5}{242}$$

(2) 在 $x=0.15$ 函数值为 $\int_0^{0.15} \frac{1}{\beta(6,5)} x^5 (1-x)^4 = 0.0014$

32. $f_{X(2), X(4)}(x, y) = \frac{5!}{1!1!1!} \cdot x^2 (y^3 - x^3) (1-y^3) \cdot 3x^2 \cdot 3y^2$
 $= 1080 x^5 y^2 (y^3 - x^3) (1-y^3) \quad 0 < x \leq y < 1$

$f_{\frac{X(2)}{X(4)}, X(4)}(u, v) = f(uv, v) v = 1080 u^5 (1-u^3) v^4 (1-v^3)$

可分离成 u 的函数与 v 的函数

$\therefore \frac{X(2)}{X(4)} \text{ 与 } X(4) \text{ 相互独立}$